

Williard Joshua Jose

williard.xyz || [linkedin.com/in/williard-jose](https://www.linkedin.com/in/williard-jose)

EDUCATION

University of Massachusetts Amherst, MA, USA

MS/PhD in Computer Science (GPA: 4.00/4.00), September 2022 – May 2027 (expected) [Adviser: Hao Zhang]

Working on multi-robot collaboration and adaptation using reinforcement learning, diffusion policy, and flow matching

University of the Philippines, Quezon City, Philippines

Master of Science in Electrical Engineering (GPA: 3.75/4.00)

Bachelor of Science in Electronics and Communications Engineering (GPA: 3.91/4.00)

Summa cum laude and class valedictorian (rank 1) of 3,000+ graduating students

PUBLICATIONS ([Link to Google Scholar](#))

Adapting Execution-Time Objectives for Multi-Robot Policies via Collaborative Flow Policy Guidance [\[Paper\]](#) [\[Page\]](#)

[Williard Joshua Jose](#), Yuhao Li, Zihao Deng, Hao Zhang. *Robotics: Science and Systems (RSS)*, 2026

Collaborative Quadraped Transportation in 3D Terrain with Constrained Diffusion [\[Paper\]](#) [\[Page\]](#)

[Williard Joshua Jose](#), Li Chen, Hao Zhang. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026

Subteaming and Adaptive Formation Control for Coordinated Multi-Robot Navigation [\[Paper\]](#) [\[Page\]](#)

Zihao Deng, Peng Gao, [Williard Joshua Jose](#), Maggie Wigness, John Rogers III, Brian Reilly, Christopher Reardon, Hao Zhang. *Conference on Robot Learning (CoRL)*, 2025

Coordinated Multi-Robot Navigation with Formation Adaptation [\[Paper\]](#) [\[Page\]](#)

Zihao Deng, Peng Gao, [Williard Joshua Jose](#), Christopher Reardon, Maggie Wigness, John Rogers III, Hao Zhang. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025

Bandwidth-Adaptive Spatiotemporal Correspondence Identification for Collaborative Perception [\[Paper\]](#) [\[Page\]](#)

Peng Gao, [Williard Joshua Jose](#), Hao Zhang. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025

Learning for Dynamic Subteaming and Voluntary Waiting in Heterogeneous Multi-Robot Collaborative Scheduling [\[Paper\]](#) [\[Page\]](#) *Finalist for Best Paper Award on Multi-Robot Systems*

[Williard Joshua Jose](#), Hao Zhang. *IEEE International Conference on Robotics and Automation (ICRA)*, 2024

AMRConvNet: AMR-Coded Speech Enhancement Using Convolutional Neural Networks [\[Paper\]](#)

[Williard Joshua Jose](#). *IEEE International Conference on Systems, Man and Cybernetics (SMC)*, 2020

Development of a 3D-Printed Biped Robot with Distributed Joint Control [\[Paper\]](#)

[Williard Joshua Jose](#), Manuel Ramos, Jr. *IEEE International Conference on Control, Automation and Robotics (ICCAR)*, 2018

PRE-PRINTS

Harmonizing Team and Individual Goals in Decentralized Multi-Robot Fire Suppression through Curriculum Reinforcement Learning [\[Paper\]](#) [\[Page\]](#)

[Williard Joshua Jose](#), Yuhao Li, Hao Zhang. *In submission*

EdgeGrasper: Fast Polygon Segmentation-Based Grasp Detection [\[Paper\]](#) [\[Page\]](#)

[Williard Joshua Jose](#), Bertram David Matabang, Joemer Aliman, Michel Onasis Ogbinar, Rowel Atienza. *In submission*

Bilevel Learning for Dual-Quadraped Collaborative Transportation under Kinematic and Anisotropic Velocity Constraints [\[Paper\]](#) [\[Page\]](#)

[Williard Joshua Jose](#), Hao Zhang. *In submission*

WORK EXPERIENCE

Distributed Embodied AI Systems Intern, Bosch Research, June 2026 – Present

Adapting generative robot policies (VLA/diffusion/flow policies) for loco-manipulation (Unitree B1/Z1) while addressing real-world communication challenges (e.g., latency, packet drops) during closed-loop execution from the cloud

Graduate Research and Teaching Assistant, University of Massachusetts Amherst, September 2022 – Present

Multi-robot collaboration: learn robot team-based strategies for collaborative transportation and formation control (DARPA)

Adaptive collaborative robot perception: develop ML methods for bandwidth-limited co-perception (Toyota InfoTech Labs)

Teaching assistant for the graduate course CS 603: Robotics

Lead Engineer (AI), Samsung R&D Institute Philippines, June 2018 – June 2022 (*on study leave*)

Led a team of 3 engineers on ML projects in 5G networks for resource optimization, information retrieval, anomaly detection

Performed R&D on deep learning, computer vision, and speech, resulting in patents (6 granted, 1 pending) and 1 publication

Lecturer 2, University of the Philippines, February 2022 – August 2022

Taught undergraduate engineering students for EEE 121: Data Structures and Algorithms

Management Associate, Globe Telecom, September 2017 – May 2018

Rotated in multiple roles: network planning of data services platforms, product management of internet access products, and project management of an enterprise transformation project

Student Assistant, University of the Philippines, August 2015 – May 2016

Performed system administration and mass software and virtual machine deployment to computer classrooms

Intern, Maxim Integrated (*acquired by Analog Devices*), June 2015 – July 2015

Debugged and modified hardware and software test systems for IC testing as an intern under the Test Optimization Team

HONORS & AWARDS

Robert and Deanna Hagerty Robotics Scholarship, University of Massachusetts Amherst, MA, USA, September 2024

Competitive scholarship awarded to outstanding students for their excellence in their research in the area of robotics

Finalist for Best Paper Award on Multi-Robot Systems, Yokohama, Japan, May 2024

IEEE International Conference on Robotics and Automation

Travel Award Grant, Conference on Robot Learning (CoRL), Seoul, South Korea, November 2024

Travel Award Grant, IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024

Robin Popplestone Memorial Fellowship, University of Massachusetts Amherst, MA, USA, May 2023

Competitive fellowship awarded to a Computer Science graduate student pursuing research in Robotics/Artificial Intelligence

Employee of the Year, Samsung R&D Institute Philippines, 2020

Only 1 employee is awarded annually, demonstrating excellence and delivering significant value for the research center

Samsung SW Certificate Expert, Samsung Electronics, 2019

Highest certification level; less than 1% of Samsung engineers globally are Expert level in algorithms and data structures

1st Place, 12th Samsung Hackathon (250 Global Participants), Samsung C-Lab, Seoul, South Korea, November 2018

Annual open innovation event where our team beat entries from Samsung (HQ-Korea, Poland, China, India, Ukraine, Russia) and top university students in Korea

Top 2, Board of Electronics Engineers, Professional Regulation Commission, Philippines, October 2017

2nd highest out of 4,784 examinees of the national licensure exam for electronics engineers

World Finalist, ACM International Collegiate Programming Contest, Rapid City, SD, USA, May 2017

The ACM-ICPC is the world's oldest and most prestigious programming contest; seeded first place in the Manila regional site

Chancellor Award for Student Distinction, University of the Philippines Diliman, Philippines, April 2017

1 of 4 annual awardees out of 24,000+ students; given for all-around excellence in academics, leadership, and extracurriculars

Oblation Scholar, University of the Philippines, Philippines, 2012 – 2017

Prestigious fully funded scholarship awarded to the top 50 out of 60,000+ university applicants

DOST-SEI Merit Scholar, University of the Philippines, Philippines, 2012 – 2017

Competitive scholarship awarded to students with high aptitude in science and mathematics

Gold Medalist, International Earth Science Olympiad, Modena, Italy, September 2011

Silver Medalist, International Junior Science Olympiad, Abuja, Nigeria, December 2010

PATENTS

Device and Method for Wirelessly Communicating on Basis of Neural Network Model, US11508394B2 (2022),

KR102645659B1 (2024), PH12019050079 (2025), EP3843092A1 (pending). Inventors: Williard Joshua Jose

System and Method for Bidirectional Automatic Sign Language Translation and Production, US12437673B2 (2025),

PH12022050141 (2025). Inventors: Daryl Peralta, Shakira Arguelles, Williard Joshua Jose

Method and Article of Manufacture for Providing a Mobility Management with Optimized Paging, US12262344B2 (2025),

PH12021050520 (2026). Inventors: Luis Gabriel Del Rosario, Amielle Dulay, Williard Joshua Jose, John William Orillo,

Joseph Alan Baking

System and Method for Intent-Based Orchestration in Radio Access Networks, PH12022050117 (2025). Inventors: Amielle

Dulay, Williard Joshua Jose

System of Calibrating Device Configurations Using Room Geometry, PH12021050201 (2024). Inventors: Shakira Arguelles,

Alfonzo Azucenas, Harjit Singh Brar, Cyril Josef Cortez, Williard Joshua Jose, James Carl Necio, Jose Ricardo IV Ruiz

System and Method for Time-of-Flight Depth Map Enhancement with RGB Images, PH12022050126 (2026). Inventors:

Williard Joshua Jose, John William Orillo, Areeya Rubenecia, Juan Raphael Seña

Method for Media Enhancement and Bandwidth Optimization on Communication Networks, PH12019050293 (pending).

Inventors: Williard Joshua Jose

PROFESSIONAL ACTIVITIES

Reviewer for ICRA (2025-2026), CoRL (2025-2026), RA-L (2024-2026), IROS (2024-2026), CASE (2025), SMC (2020, 2025)

Student Volunteer, ICCV 2023, Paris, France

PhD Application Reviewer 2025-2026, University of Massachusetts Amherst

Mentor, Undergraduate Research Volunteers (URV) 2024, University of Massachusetts Amherst
Mentor, Early Research Scholars Program (ERSP) 2023-2024, University of Massachusetts Amherst
Volunteer, PhD Applicant Support Program (PASP) 2023, University of Massachusetts Amherst

PROFESSIONAL ORGANIZATIONS

Graduate Student Member, Institute of Electrical and Electronics Engineers (IEEE), 2016 – Present
Graduate Student Member, IEEE Robotics and Automation Society (IEEE RAS), 2016 – Present
Student Member, Association for the Advancement of Artificial Intelligence (AAAI), 2026
Associate Member, National Research Council of the Philippines (NRCP), 2022 – Present
Member, Institute of Electronics Engineers of the Philippines (IECEP), 2017 – Present
Member, Mechatronics and Robotics Society of the Philippines (MRSP), 2026 – Present

SKILLS

Programming Languages: Python, C/C++, Java, MATLAB
Machine Learning Frameworks: PyTorch, TorchRL, PyTorch Geometric, TensorFlow, Jax, NumPy, Pandas, Scikit-learn, SciPy
Robotics Frameworks: ROS 1 (Melodic/Noetic), ROS 2 (Foxy/Humble/Jazzy)
Simulators: Genesis, Isaac Lab/Isaac Sim, Gazebo, MuJoCo, PyGame, VMAS
System Administration: Bash, Slurm, Docker, Kubernetes, Cloud AI/ML and data certifications ([AWS](#), [GCP](#), [Azure](#))